

Bayesian Quadrature

This section will introduce Bayesian quadrature as a practical tool for numerical integration, and will give the key procedures needed to apply it to real problems.

We will tackle the generic integration problem of computing

$$F = \int_{\mathcal{X}} f(x) \, d\nu(x), \quad (10.1)$$

where $\nu(x)$ is a measure.¹ The domain of integration, $\mathcal{X}$, is, in practical applications, often a bounded interval, such as $[a, b] \subset \mathbb{R}$. The integration problem (10.1) is written as univariate, which will be the setting motivating the majority of this chapter. We will, however, also explain in §10.1.2 how Bayesian quadrature can be generalised to multivariate problems.

As we already saw in Chapter I, Gaussian process models for f are a fundamental tool for efficient formulations of inference. And indeed they allow for an analytic formulation of integration as probabilistic inference. Because Gaussian measures are closed under linear maps (see Eq. (3.4)), they are, in particular, also closed under integration. Following the exposition of §4.4, a generic Gaussian process prior $p(f) = \mathcal{GP}(f; m, k)$ over the integrand amounts to a joint Gaussian measure over both the function values collected in $Y := [f(x_1), \dots, f(x_N)]$ and the integral F :

$$\begin{aligned} p(F, Y) &= \int p(F, Y \mid f) p(f) \, df = \int \delta(F - \int_{\mathcal{X}} f(x) \, d\nu(x)) \prod_{i=1}^N \delta(y_i - f(x_i)) p(f) \, df \\ &= \mathcal{N} \left(\begin{bmatrix} Y \\ F \end{bmatrix}; \begin{bmatrix} m_X \\ \int_{\mathcal{X}} m(x) \, d\nu(x) \end{bmatrix}, \begin{bmatrix} k_{XX} & \int_{\mathcal{X}} k_{Xx} \, d\nu(x) \\ \int_{\mathcal{X}} k_{xx'} \, d\nu(x) \, d\nu(x') \end{bmatrix} \right). \end{aligned}$$

Here we have used the Dirac point measure δ as a likelihood term to encode *exact*, perfectly certain, observations of f at

¹ The example of Eq. (9.2) can either be seen as integrating the function $f(x) = e^{-\sin^2(3x)}$ against the Gaussian measure $\nu(x) = e^{-x^2}$, or as integrating the f from Eq. (9.1) against the Lebesgue measure.

the locations X . The conditional for F given Y , the posterior $p(F | Y)$, is a univariate Gaussian distribution² with parameters $\mathbf{m} \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}_+$, given by

$$p(F | Y) = \mathcal{N}(F; \mathbf{m}, \mathbf{v}), \quad (10.2)$$

where

$$\mathbf{m} := \int_{\mathcal{X}} m_x + k_{xX} k_{XX}^{-1} (Y - m_X) \, d\nu(x) \quad (10.3)$$

$$= \mathbf{m}_0 + \mathfrak{k}_X^\top k_{XX}^{-1} (Y - m_X),$$

$$\begin{aligned} \mathbf{v} &:= \iint_{\mathcal{X}} k_{xx'} - k_{xX} k_{XX}^{-1} k_{Xx'} \, d\nu(x) d\nu(x') \\ &= \mathfrak{K} - \mathfrak{k}_X^\top k_{XX}^{-1} \mathfrak{k}_X. \end{aligned} \quad (10.4)$$

These expressions are tractable if there are analytic forms for the following three univariate and bivariate integrals:³

$$\mathbf{m}_0 := \int_{\mathcal{X}} m(x) \, d\nu(x) \in \mathbb{R}, \quad (10.5)$$

$$\mathfrak{k}(x_i) := \int_{\mathcal{X}} k(x, x_i) \, d\nu(x) \in \mathbb{R}, \quad (10.6)$$

$$\mathfrak{K} := \iint_{\mathcal{X}} k(x, x') \, d\nu(x) d\nu(x') \in \mathbb{R}_+. \quad (10.7)$$

When these expressions are available, Eq. (10.2) amounts to an analytic map of the observations Y onto a probability measure over F with support on the entirety of, or a subset of, the real numbers. These integrals do not involve the integrand f , but the kernel k (and m) instead. Embedding the single integrand in a hypothesis class of many possible true functions (the support of the prior) in this way thus turns the intractable problem of exact integration into tractable, uncertain inference.

Notably, the posterior mean in Eq. (10.2) can be written as an affine function of Y :

$$\mathbf{m} = F_0 + w^\top Y = F_0 + \sum_{i=1}^N w_i y_i, \quad (10.8)$$

$$\text{with offset} \quad F_0 := \mathbf{m}_0 - \mathfrak{k}_X^\top k_{XX}^{-1} m_X,$$

$$\text{and weights} \quad w_i := \sum_{j=1}^N \mathfrak{k}_{x_j} [k_{XX}^{-1}]_{ji}, \quad i = 1, \dots, N.$$

The construction so far just provides a probability measure over F without making any claims on its usefulness. But we would like to interpret that measure as a probabilistic *estimate*, of F , treating the posterior mean $\mathbf{m}$ as a point estimator for F , and the posterior variance $\mathbf{v}$ as an estimate of its square error, a notion of uncertainty. Not all Gaussian process priors will be equally

² Eq. (10.2) can also be reached by first forming the posterior $p(f | Y) =$

$$\mathcal{GP}(f; m_x + k_{xX} k_{XX}^{-1} (y - m_X), \\ k_{xx'} - k_{xX} k_{XX}^{-1} k_{Xx'}),$$

and using the closure of this Gaussian process under the (linear) integration operation.

³ In practical application ν is often a probability measure. For example, in probabilistic statistics and machine learning, the typical situation is that $\nu(x)$ is a prior measure on x , while $f(x)$ is a likelihood function arising from some data. In such settings $\mathfrak{k}(x)$ equals the *kernel mean embedding* of ν in the RKHS of k , and Bayesian quadrature is related to the notion of *kernel herding* from the kernel community in machine learning. For more, see Smola et al. (2007) and Huszar and Duvenaud (2012).

“good” for a particular integration problem. To argue for one prior over another, we thus need to show that either quantity has analytic properties supporting this interpretation of point estimate and error estimate. This analysis will be provided, for a specific model choice, in §11.1.5.

► 10.1 Models

After the abstract introduction, we turn to describe tangible models for Bayesian quadrature. By default, we consider the real domain, $\mathcal{X} = \mathbb{R}$, although more general cases will also be presented.

Given a fixed set of nodes, integration requires two choices: that of the measure $\nu(x)$ and the model for the integrand. The measure $\nu(x)$ is often stipulated by the problem. It may also be possible to choose $\nu(x)$ to ease the resulting integration problem. For example, given a real domain, a Gaussian ν may be taken for the convenience of the resulting numerical integration problem.

We can now turn to the choice of model. Within Bayesian quadrature, the choice of GP prior for the integrand leads to two further choices: that of the prior mean $m(x)$ and the prior covariance $k(x, x')$. As above, these choices are not free, but have to be made such that the integrals in Eqs. (10.5)–(10.7) remain tractable. The mean m represents an initial guess for the integrand, while the covariance should aim to capture both the deviation of the true integrand from m and knowledge about analytic structure of the integrand beyond that represented by m e.g. periodicity or differentiability.

In practice, the prior mean function m is almost always taken as zero: $m(x) = 0$. It is rare that an integrand is known sufficiently well ahead of its evaluation to support any better-informed choice. The zero mean function has the additional practical advantage of rendering m_0 from Eq. (10.5) trivially zero, and thereby simplifying the posterior mean equation for the integral, Eq. (10.3).

As in §6, this chapter will exclusively use a *scaled* covariance family, separating a simple *scale* $\theta \in \mathbb{R}$ from a “unit” kernel $\tilde{k}(x, x')$,

$$k(x, x') := \theta^2 \tilde{k}(x, x'). \quad (10.9)$$

The unit kernel $\tilde{k}$ captures more intricate structure, and will control the shape of the posterior and thus the *rate* at which the uncertainty contracts. The scale θ can then be inferred hierarchically as introduced in §6, to find the scale constant. That is because the scale θ also shows up multiplicatively in the

posterior on F :

$$p(F \mid Y) = \mathcal{N}\left(F; \int_{\mathcal{X}} m_x + \tilde{k}_{xX} \tilde{k}_{XX}^{-1} (Y - m_X) dx, \theta^2 \iint_{\mathcal{X}} \tilde{k}_{xx'} - \tilde{k}_{xX} \tilde{k}_{XX}^{-1} \tilde{k}_{Xx'} dx dx'\right).$$

▷ 10.1.1 Gaussian Covariance / Squared-Exponential Kernel

Within this family of covariances is the Gaussian covariance discussed in §4.4, which can also be written indirectly using the Gaussian density function as

$$k(x, x') = \theta^2 \mathcal{N}(x; x', \lambda^2).$$

When paired with the Gaussian measure

$$\nu(x) = \mathcal{N}(x; \mu, \sigma^2),$$

it yields tractable results (listed in §4.4) for the integrals required for Bayesian quadrature (Eqs. (10.5)–(10.7)). For convenience, we repeat that the posterior for the integral F given evaluations $Y = [f(x_1), \dots, f(x_N)]$ is

$$p(F \mid Y) = \mathcal{N}(F; \mathbf{m}, \mathbf{v}),$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{f}_X^\top k_{XX}^{-1} Y, \\ \mathbf{v} &= \mathbf{K} - \mathbf{f}_X^\top k_{XX}^{-1} \mathbf{f}_X. \end{aligned}$$

For the pairing of a Gaussian measure, zero prior mean and Gaussian covariance, we arrive at

$$\begin{aligned} [\mathbf{f}]_i &= \theta^2 \mathcal{N}(x_i; \mu, \lambda^2 + \sigma^2) \in \mathbb{R}, \quad \text{for } i \in \{1, \dots, N\}, \\ \mathbf{K} &= \frac{\theta^2}{\sqrt{2\pi(2\sigma^2 + \lambda^2)}} \in \mathbb{R}. \end{aligned}$$

Now we have all the ingredients for a practical Bayesian quadrature procedure for univariate problems.

▷ 10.1.2 Multivariate Integrals

Multivariate integrals are common. Fortunately, Bayesian quadrature is, in principle, readily extended to integrals over d variables (which we compile into the d -dimensional vector x). This extension requires only the choice of a Gaussian process prior over the d -dimensional space that yields tractable integrals against the

measure. For instance, if the domain is $\mathbb{R}^d$, with multi-variate Gaussian measure

$$\nu(x) = \mathcal{N}(x; \mu, \Sigma),$$

and we take a GP with zero prior mean and multi-dimensional Gaussian covariance

$$k(x, x') = \theta^2 \mathcal{N}(x; x', \Lambda), \quad (10.10)$$

we arrive at (where x_i is the i th node, a d -dimensional vector)

$$\begin{aligned} [\mathfrak{k}]_i &= \theta^2 \mathcal{N}(x_i; \mu, \Lambda + \Sigma) \in \mathbb{R}, \quad \text{for } i \in \{1, \dots, N\}, \\ \mathfrak{K} &= \theta^2 (\det 2\pi(2\Sigma + \Lambda))^{-\frac{1}{2}} \in \mathbb{R}. \end{aligned}$$

In practice, however, such a simple BQ model is unlikely to work well in high dimension. In general, multivariate integration is plagued by the “curse of dimensionality”: the volume to be integrated over grows exponentially in dimension. As a consequence, if the number of evaluations does not grow similarly exponentially, a high-dimensional volume will be only sparsely explored by the evaluations. As a result, the higher the dimension, the greater the importance of the model, and the lesser the importance of evaluations of the integrand. The Gaussian covariance (10.10) is particularly poor in high dimension. The light tails of the Gaussian covariance prevent an evaluation from having influence over GP predictions at far-removed locations: exactly what is necessary in order to predict anything other than the GP prior mean in sparsely-explored spaces. At the time of writing, Bayesian quadrature is thus rarely used in dimension greater than around 20.

▷ 10.1.3 Other Tractable Models

We conclude this section with a survey of tractable models for Bayesian quadrature. Table 10.1 reviews other pairings of prior measure $\nu(x)$ and Gaussian process kernel $k(x, x')$ that yield tractable results for the Bayesian quadrature equations, Eqs. (10.5)–(10.7). All assume a zero prior mean. This table is, by no means, to be considered exhaustive: a creative mind may be able to frame a given problem in additional ways to yield computable results. In doing so, it may be guided by the concrete interpretation of the model choice as a prior on functions. For example, one may consider the properties of samples from the prior to gain intuition for their fit to a particular problem. Classic quadrature methods also form a similar “zoo” of choices

$\mathcal{X}$	ν	k	Reference
$[0, 1]^d$	Unif($\mathcal{X}$)	Wendland TP	Oates et al. (2019a)
$[0, 1]^d$	Unif($\mathcal{X}$)	Matérn Weighted TP	Briol et al. (2019)
$[0, 1]^d$	Unif($\mathcal{X}$)	Gaussian	Use of error function
$\mathbb{R}^d$	Mix. of Gaussians	Gaussian	Kennedy (1998)
$\mathbb{S}^d$	Unif($\mathcal{X}$)	Gegenbauer	Briol et al. (2019)
Arbitrary	Unif($\mathcal{X}$) or mix. of Gauss.	Trigonometric	Integration by parts
Arbitrary	Unif($\mathcal{X}$)	Splines	Wahba (1990)
Arbitrary	Known moments	Polynomial TP	Briol et al. (2015)
Arbitrary	Known $\partial \log \nu(x)$	Gradient-based kernel	Oates, Girolami, and Chopin (2017); Oates et al. (2019a)

(see Table 11.1), but choosing between them is a less intuitive process, and adding to them is even harder, because there are only analytical, rather than constructive, means to do so without the interpretation of a prior. Interpretability is a key strength of PN.

► 10.2 Node Selection

The Bayesian quadrature model engenders a natural design rule. That is, thinking of the design rule as resulting from the minimisation of an expected loss, one loss function immediately suggests itself: the square error. The expected loss is then equal to the variance of $p(F | Y)$, v (the expected square error after receiving the evaluation Y). The resulting design rule selects the locations $X = [x_1, \dots, x_N]$ such that v is minimised (note that v is a function of X , a fact we will make explicit here):⁴

$$X = \arg \min_{\tilde{X} \in \mathbb{R}^N} v(\tilde{X}).$$

Designing the optimal grid for such a rule, even for regular kernels, can be challenging, because the corresponding multivariate optimisation problem can, in general, have high computational complexity.⁵ However, instead of finding an *optimal* grid, one can also *sample*, at cost $\mathcal{O}(N^3)$, a draw from the *N-determinantal point process* (DPP) associated with k . Results by Bardenet and Hardy (2019) suggest that doing so causes only limited decrease in performance over the optimal deterministic grid design (which amounts to the maximum a posteriori assignment under the *N-DPP*). Belhadji, Bardenet, and Chainais (2019) offer further support for the use of determinantal point process sampling for integrands known to live in an RKHS.⁶

Table 10.1: A non-exhaustive list of distribution ν and kernel k pairs that provide a closed-form expression for both the kernel mean $\int k(\cdot, x) dx$ and the initial error $\mathfrak{R}$ from Eq. (10.7). Here TP refers to the tensor product of one-dimensional kernels, and $\mathbb{S}^d$ indicates the d -sphere $\{x = (x_1, \dots, x_{d+1}) \in \mathbb{R}^{d+1} : \|x\|_2 = 1\}$. This table is adapted from Briol et al. (2019).

⁴ Because the entropy of a Gaussian is a monotonic function of its variance (3.3), this design rule can also be phrased in an information-theoretic way as choosing the “most informative” design – that which minimises the conditional entropy of F given Y .

⁵ Ko, Lee, and Queyranne (1995)

⁶ The point of these papers does not disagree with our general argument, in §12.3, against the use of random sampling. Rather, these results show that allowing minor deviations from the optimal design can drastically reduce computational complexity at negligible decrease in performance. The necessary “samples” can even be drawn in a deterministic way.

Recall from Eq. (10.4) that the variance, v , of the integral does not depend on the values of the observations, Y , at all. Hence this is an open-loop, non-adaptive design criterion. This is simultaneously a good and a bad property of all Gaussian models. It means an evaluation pattern can be chosen a priori and does not need to be adapted at runtime, potentially saving computation time. However, the independence of the variance from Y also implies that the error estimate (and the evaluation locations) will be based entirely on prior assumptions, and will not depend on the collected function values themselves. These facts may feel disquieting: should we not hope that the evaluations inform both our confidence and the locations of future evaluations? To address these concerns, adaptive Bayesian quadrature schemes have been devised.

▷ 10.2.1 Adaptive Selection

Adaptive Bayesian quadrature schemes have, broadly, adopted non-Gaussian models, thereby shedding both the good and bad of such models. To motivate such models, note that many integration problems possess non-negative integrands. The classic case is probabilistic inference, where many integrands are a product of non-negative densities like likelihoods and priors. A particularly common case is the model evidence (or marginal likelihood),

$$\underbrace{p(\mathcal{D})}_F = \int \underbrace{p(\mathcal{D} \mid x)}_{f(x)} \underbrace{p(x)}_{dv(x)} dx. \quad (10.11)$$

As noted in the motivation for this chapter, §9.1, solving such integrals may also be a key step towards AI. To date, all known Bayesian quadrature schemes that are adaptive (according to our definition in §9.3) consider this setting of non-negative integrands.

The first attempt at adaptive Bayesian quadrature⁷ was termed *doubly-Bayesian quadrature* (BBQ) for the (Bayesian) decision-theoretic treatment of Bayesian quadrature. BBQ adopts an (approximate) means of modelling the logarithm of the integrand with a GP, thereby (approximately) enforcing the non-negativity of the integrand. The motivation for this model is the large dynamic range (variation over many orders of magnitude) of many probabilistic integrands: a GP on the logarithm allows the model to learn such a large dynamic range. To select nodes, BBQ uses the variance of the integral, (10.4), as a loss function, so that the expected loss is the expected square-error after evaluating at that node. As such, BBQ uses what is arguably both

⁷ Osborne et al. (2012)

the most desirable model (a log-GP) and the most desirable loss function. However, **BBQ** employs a first-order approximation to the exponential function, along with the maintenance of a set of *candidate points* x_c at which to refine the approximation. Such approximation proves both highly computationally demanding and to express only weakly the prior knowledge of the large dynamic range of the integrand.

The first practical adaptive Bayesian quadrature algorithm was **wsabi**,⁸ which adopts another means of expressing the non-negativity of an integrand: the square-root of the integrand $f(x)$ (minus a constant, $\alpha \in \mathbb{R}$) is modelled with a GP. Precisely,

⁸ Gunter et al. (2014)

$$f(x) = \alpha + \frac{1}{2}\tilde{f}(x)^2,$$

where, given data $\mathcal{D}$,

$$p(\tilde{f} \mid \mathcal{D}) = \mathcal{GP}(\tilde{f}; \tilde{m}, \tilde{\mathbb{V}}),$$

where $\tilde{m}(x)$ and $\tilde{\mathbb{V}}(x, x')$ are the usual GP posterior mean (4.6) and covariance (4.7), respectively (and thus depend on $\mathcal{D}$). An integrand modelled as the square of a GP will have a smaller dynamic range than one modelled as an exponentiated GP. In this respect, **wsabi** is a step backwards from **BBQ**.

However, the **BBQ** approximations are significantly more costly, both in computation and quality, than those required for **wsabi**. That is, **wsabi** considers both linearisation and moment-matched approximation to implement the square-transformation: both prove more tractable than the linearised-exponential for **BBQ**. Linearisation gives the following (approximate) posterior for the integrand:

$$\begin{aligned} p(f \mid \mathcal{D}) &\simeq \mathcal{GP}(f; m^{\mathcal{L}}, \mathbb{V}^{\mathcal{L}}), \\ m^{\mathcal{L}}(x) &:= \alpha + \frac{1}{2}\tilde{m}(x)^2, \\ \mathbb{V}^{\mathcal{L}}(x, x') &:= \tilde{m}(x)\tilde{\mathbb{V}}(x, x')\tilde{m}(x'). \end{aligned} \quad (10.12)$$

The posterior for moment-matching is

$$\begin{aligned} p(f \mid \mathcal{D}) &\simeq \mathcal{GP}(f; m^{\mathcal{M}}, \mathbb{V}^{\mathcal{M}}), \\ m^{\mathcal{M}}(x) &:= \alpha + \frac{1}{2}(\tilde{m}(x)^2 + \tilde{\mathbb{V}}(x, x)), \\ \mathbb{V}^{\mathcal{M}}(x, x') &:= \frac{1}{2}\tilde{\mathbb{V}}(x, x')^2 + \tilde{m}(x)\tilde{\mathbb{V}}(x, x')\tilde{m}(x'). \end{aligned} \quad (10.13)$$

The expressions above demonstrate that both linearised and moment-matched are readily implemented. In either case, the posterior for the integrand is a GP, manipulable using the standard GP equations (e.g. Eq. (4.6) for the posterior mean and

Eq. (4.7) for the covariance). A closely related approach, *moment-matched log transformation* (MMLT),⁹ instead proposed moment-matching for the (arguably more desirable) exponential transformation, which, again, proves less costly than BBQ.

As a loss function, both WSABI and MMLT use the (point-wise) variance in the *integrand* (rather than the variance in the *integral*, as per BBQ). This policy, of sampling a function where its variance is highest, is known generically as *uncertainty sampling*. That is, uncertainty sampling is the sequential design rule

$$x_{i+1} = \arg \max_x \mathbb{V}(x, x),$$

where $\mathbb{V}(x, x)$ is the posterior variance for the integrand (that is, $\mathbb{V}^{\mathcal{L}}(x, x)$, Eq. (10.12), for linearised WSABI and $\mathbb{V}^{\mathcal{M}}(x, x)$, Eq. (10.13), for moment-matched WSABI). Uncertainty sampling seems sensible: by reducing the point-wise variance in the integrand, we are presumably also reducing the variance in the integral so that, in the limit of evaluations, we should expect convergence of the estimate for the integral. However, uncertainty sampling is less clearly related to the ultimate goal of quadrature than targeting the variance (and hence squared error) of the integral.

For instance,¹⁰ consider the integral

$$F = \int_0^{2\pi} f(x) dx,$$

where $f(x) := \sin(x + \phi)$ is known, but the phase $\phi \in \mathbb{R}$ is unknown. As a result, the pointwise variance in the integrand, $\text{var}(f(x))$, will depend on $p(\phi)$, and will generally be non-zero. If our loss function is the pointwise variance, we will be impelled to take evaluations (potentially many, if the evaluations are noisy) – however, these evaluations have no impact on our belief in F . Conversely, if our loss function is the variance in the integral, we need not make any evaluations at all. Given just prior information, we arrive at $p(F) = \delta(F)$: we can be certain that $F = 0$, so $\text{var}(F) = 0$. In such, arguably corner cases, uncertainty sampling is clearly wasteful.

However, in practice, uncertainty sampling leads to algorithms that are usually substantially faster than BBQ. In fact, both MMLT and WSABI have demonstrated convergence that is significantly faster than Monte Carlo competitors – not in the number of evaluations, but in *wall-clock time*. As examples, Figures 10.1 and 10.2 demonstrate that WSABI can achieve a given estimation error in less time (measured in seconds) than even a sophisticated competitor like *annealed importance sampling*

⁹ Chai and Garnett (2019)

¹⁰ Thanks to Roman Garnett for this example.

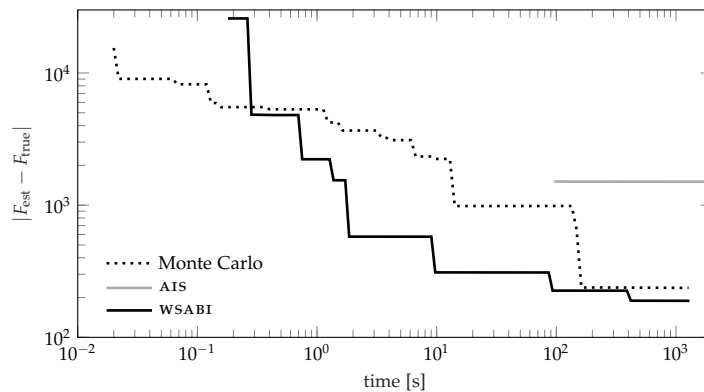


Figure 10.1: A comparison of approaches to estimating the marginal likelihood F from Eq. (10.11) for a GP regressor fitted to real data – the yacht hydrodynamics benchmark data set (Gerritsma, Onnink, and Versluis, 1981). The required integral was eight-dimensional. This plot is adapted from Gunter et al. (2014), which contains full details.

(AIS).¹¹ These are remarkable results, because the algorithm for `wsabi` requires the maintenance of a full GP, with its costly $\mathcal{O}(N^3)$ computational scaling in N , the number of evaluations. These algorithms also require the computationally expensive management of the GP's hyperparameters. `wsabi` also selects evaluations at each iteration by solving a global optimisation problem, making calls to optimisation algorithms. In short, `wsabi` requires substantial computational overhead. In comparison, Monte Carlo makes use of a (*pseudo-*) random number generator (PRNG) to select evaluations, at negligible overhead cost, and a negligibly costly simple average as a model. Nonetheless, the substantial overhead incurred by `wsabi`, included in those measurements of wall-clock time, does not prevent `wsabi` from converging more quickly than Monte Carlo alternatives. Again, the overhead is perhaps better framed as an investment, whose returns more than compensate for the initial outlay. We will return to these considerations in our generic arguments against random numbers in §12.3.

¹¹ Neal (2001)

Buttressing these promising empirical results, Kanagawa and

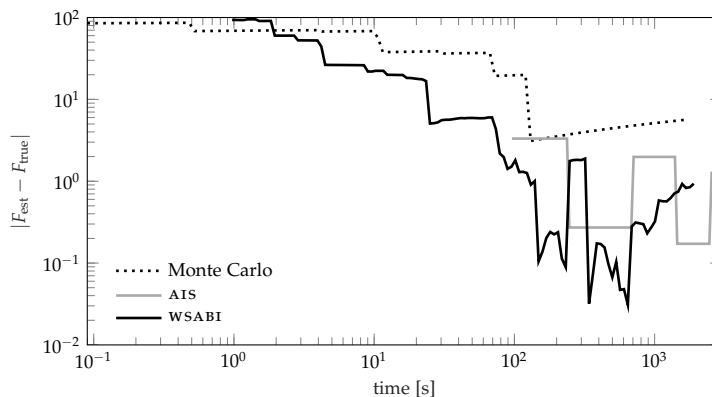


Figure 10.2: A comparison of approaches to evaluating the marginal likelihood F from Eq. (10.11) for a GP classifier fitted to real citation network data (Garnett et al., 2012). The required integral was four-dimensional. Further details are available in the plot's original source (Gunter et al., 2014).

Hennig (2019) provide a theoretical framework for adaptive Bayesian quadrature schemes. It develops a novel property – termed *weak adaptivity* – that guarantees consistency (subject to further regularity conditions) for adaptive BQ rules with a sequential design rule given by

$$x_{i+1} = \arg \max_x a_i(x),$$

with

$$a_i(x) = T(q^2(x)\tilde{\mathbb{V}}(x, x)) b_i(x),$$

where $\tilde{\mathbb{V}}(x, x)$ is the usual GP posterior variance from Eq. (4.7), q is a positive function (e.g. the integration measure), T is the transformation (also known as warping), and $b_i(x)$ is an “adaptivity” function (in linearised *wsABI*, that is the square posterior mean). Weak adaptivity, loosely speaking, requires that b_i is bounded away from zero and infinity. Linearised *wsABI* does not technically fulfil this condition, but can be made to do so with a minor correction. Moment-matched *wsABI* and *mMLT* do satisfy the condition. Intuitively, weak adaptivity means that the method is not “too adaptive” relative to non-adaptive BQ, and so can only be “stuck for a while, but not forever”. Kana-gawa and Hennig (2019) also provide a worst-case bound on the convergence rate that *approaches* that of non-adaptive Bayesian quadrature. This result is weaker than that desired: empirically, we usually see adaptive schemes offer *improved* convergence over non-adaptive schemes.

Weak adaptivity plays a conceptually analogous role to the notions of *detailed balance* and *ergodicity* used to show that MCMC algorithms, likewise, can be “stuck for a while, but not forever”. For MCMC the resulting insight is that, on some unknown time-scale, the mixing time of the Markov Chain, the algorithm converges like direct Monte Carlo. Similarly, weak adaptivity shows that, up to some constants, adaptive BQ works at least as well as its non-adaptive counterpart. Detailed balance and ergodicity alone don’t necessarily make a good MCMC method (consistency is a very weak property, after all). However, the statistical community has used the theoretical underpinning provided by detailed balance and ergodicity as a licence to develop a diverse zoo of MCMC methods that are chiefly evaluated empirically. One may hope that the licence of weak adaptivity might enable a similar flourishing of adaptive BQ schemes.

